

Supplementary Material

Reading the Banished: A Deep-Learning Emotion Analysis of Tang–Song Exiled Literati

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This supplement carries the material referred to from the main text as *Supplementary Material*: the corpus-construction record (App. A), the cross-edition invariant-representation probe (App. B), and the lexicon, exile chronology and gold-annotation protocol (App. C). It is self-contained and cites the main paper only by name.

1 Corpus Construction: Transcription, Parsing and Attribution

Yu Xuan Song Shi survives only as the mandoku plaintext of *Yu Xuan Song Jin Yuan Ming Si Chao Shi* (kanripo KR4h0143), in which the 78 Song *juan* interleave author registers, prefaces and a handful of Jin/Ming *juan*. We parsed every *shi* body line (the lowest-indent text) and attributed it to its enclosing author by exact-matching the (simplified) author name against the *Quan Song Shi* author registry of 8,925 names; this yields 9,405 poems by 660 authors, and *all* 660 match an author present in QSS, confirming the parse. *Song Shi Chao* is transcribed the same way: verse is the lowest-indent text, each poet’s section is anchored on the *ji*-name heading—including the 上/中/下 continuation headings that carry 欧阳修, 王安石 and 苏轼 across several volumes—and the prose biographies that habitually follow those headings are discarded rather than counted as verse, since they narrate demotion and transfer and would inflate the 贬谪羁旅 and 悲苦孤寂 registers on the anthology side alone. The parse anchors 98 poet headings and yields 9,612 poems by 76 authors over 829,341 characters of verse, with no unresolved attribution.

Two documented features of this text bear on how the comparison should be read. The *Siku* digest records that sixteen of the poets it lists appear here as headings without text (“有錄無書...蓋剞劂未竣故竟無完帙也”), and describes the anthology as gathering 百家; if exact, this state should yield eighty-four poets with verse, and our 76 leave a residue of eight that we neither anchor nor explain away. The shortfall affects author *coverage* only: δ is computed over authors present in both sources and is not directly biased by it. The same digest also reports that because each poet’s section was printed as a separate booklet, copies in circulation differ in extent (“所傳之本往往多寡不同”)—the paper’s own thesis stated by a Qing editor, for “the edition” of a selection anthology is not a stable object even before measurement begins.

Attribution—identifying *who* wrote a poem—is a separate question from whether the affect measurement is trustworthy, and we keep it out of the main line. The task is well studied: Cap-Transformer models for Tang-poet identification [Zhou et al., 2022], LDA-Transformer mixtures [Zhou et al., 2021], and imagery-based style separating 柳永 from 苏轼 [Fang, 2011]. We compare three classifiers on identical 85/15 stratified splits under one fixed seed: a character CNN [Kim, 2014], character n -gram TF-IDF with logistic regression, and the same features with a linear SVM. Table 1 shows the lightweight TF-IDF + SVM *beats* the CNN on every task and is more reproducible and GPU-free; the 12-way author task is held back by severe class imbalance (minor authors 154–320 poems, major authors 2,000+), which gives low macro-F1 despite high weighted-F1.

Table 1: Attribution: accuracy / macro-F1. Best per row in bold.

Task	CNN	TF-IDF+LR	TF-IDF+SVM
Author (12)	.624/.271	.714/.351	.802/.568
Dynasty (2)	.894/.837	.895/.808	.936/.898
Tang prose (4)	.619/.512	.856/.812	.924/.908

2 Cross-Edition Invariant-Representation Probe (M6)

As a probe—not a core claim—we ask whether a *learned* representation could hold a poet’s affect while shedding edition-specific leakage. We freeze the PPMI–SVD character embeddings (100-d, unit-norm), represent each poem by the mean-pool of its character vectors, and take the author-mean vector for every author present in *both* editions with ≥ 300 characters. Affect is read out as in the transmission-robustness analysis of the main paper: project each author vector onto the five lexicon pole directions and compute (A, S) . We obtain a single linear “source-invariant” projector by a *closed-form domain-adversarial* solution—train a logistic adversary to predict edition from the author-mean vector, then project out its weight direction, $g(\mathbf{x}) = \mathbf{x} - (\mathbf{x} \cdot \mathbf{w})\mathbf{w}$ —and apply it to both settings.

Table 2: M6 invariant-representation probe. Frozen = identity projector on the PPMI–SVD author-mean backbone; M6 = closed-form domain-adversarial projector. A/S are the concordance coefficients read out from the 5-dim pole projection. edAUROC = 5-fold logistic edition predictability (lower is more invariant); varR = dimensionwise variance retained by M6 (1.0 = none removed).

Setting	n	Variant	A	S	edAUROC	varR
Tang: QTS vs. YD	363	frozen	0.865	1.00	0.536	1.00
Tang: QTS vs. YD	363	M6	0.981	1.00	0.537	0.69
Song: QSS vs. YXSS	216	frozen	0.900	1.00	0.848	1.00
Song: QSS vs. YXSS	216	M6	0.980	1.00	0.584	0.49

On the Tang same-work pair the edition direction is negligible to begin with (edAUROC $0.536 \approx \text{random}$), so M6 has almost nothing to remove. On the Song selection pair it is large and dominant (frozen edAUROC 0.848); M6 removes precisely that axis, collapsing edition predictability to 0.584 while retaining all five affect dimensions ($S=1.00$) and raising A from 0.900 to 0.980. This is a *probe*: its A/S values are *not* directly comparable to the cross-edition concordance table of the main paper, and its only licensed reading is qualitative—edition information here is a single, low-rank, separable direction removable without harming the affect dimensions, consistent with the main finding that edition effects act along specific directions rather than as diffuse noise.

3 Lexicon, Chronology and Annotation Protocol

The lexicon $\mathcal{L}$ partitions words into five categories; the *full* lists are recorded below. Each list mixes single characters with multi-character words; all entries are stored in simplified script, consistent with the corpora’s normalisation (traditional→simplified via OpenCC). The two affective poles are marked (A); the thematic exile–journey category ($A^\dagger$, the named condition rather than a felt affect) is marked separately; the two imagistic registers are marked (I).

悲苦孤寂 BEI (A; sorrow–loneliness): 悲伤愁怨苦孤寂独凄惨哀痛恨忧戚惘怆涕泪殇歿愍悴惻黯销魂零萧雁砧捣更灯店舫猿驿帆仆僮断肠凄凉寂寥孤危憔悴销魂悲秋苦辛酸辛辛酸飘零飘蓬断蓬飞蓬转蓬

贬谪羁旅 BIAN ($A^\dagger$; exile–journey, the named condition rather than a felt affect): 贬谪迁逐流羁旅客宦远荒蛮瘴殊遐徼陬裔去国天涯海角殊方投荒远谪迁客楚客南荒炎荒瘴江蛮烟

旷达闲适 KUANG (A; detached–easeful): 闲适达旷豁放醉啸傲悠淡泊乐欣悦遣兀疏狂从容恬静禅苔钓隐亭栽莲畦眠旷达闲适放达自适陶然悠然忘机委运随遇达观适意

自然山水 ZI (I; nature–landscape): 山水风月林泉石竹松江河湖海花鸟溪岩岚汀洲屿峰崖涧泽涯渚苹蓼清风明月白云沧浪烟霞苇岸渔樵桃源落英幽篁暮云孤云寒云春云秋云停云烟云云霞云峰云山

空幻时空 KONG (I; void–temporality): 空幻梦影尘虚浮瞬逝昔古千秋万须臾泡电光隙驹阴浮生如梦梦幻须臾万古古今今古泡影微尘刹那弹指过眼逆旅倏忽

Corpus coverage of the lexicon. Table 3 reports how often each category fires on the unified *shi* corpus (21,287 poems, 1,466,827 characters). All five categories fire frequently and in reasonable balance; 自然山水 is expectedly the most frequent, reflecting the centrality of landscape in classical verse.

Table 3: Lexicon corpus coverage (unified *shi* layer).

Category	Entries	Hits	Poems hit (%)	Rate / 1k
悲苦孤寂 Bei	56	18,677	10,303 (48.4%)	12.73
贬谪羁旅 Bian	30	9,549	6,721 (31.6%)	6.51
旷达闲适 Kuang	43	19,446	11,054 (51.9%)	13.26
自然山水 Zi	49	45,254	16,478 (77.4%)	30.85
空幻时空 Kong	36	26,442	13,308 (62.5%)	18.03

The ABI is $\text{ABI}(a) = \phi_{a,\text{KUANG}} - \phi_{a,\text{BEI}}$ (Eq. 2).

The quietist (静禅) register is carried by the 旷达 detached-easeful class via 禅/苔/钓/隐/亭/栽/畦/眠; the exile-journey and homesickness registers are carried by the 悲苦 sorrow-loneliness class via 驿/帆/舫/店/灯/砧/捣/仆/僮 (entry counts in `results_lexicon_v2_curation.json`: 悲苦孤寂 37→47→56, 旷达闲适 33→43).

None of the four corpora carries per-poem dating metadata (0 of 356,714 poems across QTS/QSS/YD/YXSS carries a usable year field; the *tags* of the Tang files are thematic labels and the *strains* of the Song files are tonal patterns), so within-author period assignment required an external chronology layer. For each queue official a set of exile events was compiled from the standard scholarship (biographies, chronologies, and annotated editions), each graded A (histories plus a dated chronology agree), B (scholarly consensus), or C (disputed; never merged into A/B). Poems were then attached to periods by two routes: internal chronological ordering, verified against anchor poems whose dates are certain (for Su Shi the *Quan Song Shi* sequence is chronological from its first poem; the Crow-Terrace poem, *First Arriving in Huangzhou*, *Leaving Huangzhou*, *First Arriving in Huizhou*, the Hainan poems and the last poem all fall in strict order; the supplementary block after the final poem is not chronological and was marked *unknown*), or poem-level attachment from dated annotated editions where the collection is ordered by genre rather than by year. A ± 1 -year buffer around each event boundary is excluded from all contrasts. Coverage: Su Shi 473 exile / 995 pre / 860 post poems (85% of his corpus attached at grade A); Bai Juyi 59 / 66 / 18 at grade B with a documented genre-selection caveat; the other four officials fell below pairing power on attachment (Liu Yuxi 42 exile poems with no separable pre/post control; Ouyang Xiu 12; Wang Yucheng 29; Huang Tingjian 4). Attachments, grades and coverage are released in the replication package (`poem_period_labels.csv`, `exile_chronology.csv`).

Sampling. The gold sample is 160 *shi* poems, 80 per dynasty, drawn by stratified random sampling (seed 20260917) over the cross of dynasty, membership in the twelve-poet cohort, and whether the poem’s lexicon ABI falls above or below the median; within each dynasty 60 poems come from the complete collection and 20 from a selection anthology, so that the sample spans both transmission regimes and the full range of the instrument’s output rather than only its middle. Poems are restricted to 20–160 characters; below 20 characters a per-1,000-character rate is meaningless, and above 160 characters the burden of close reading per item makes a blind

double coding impractical. The realised sample covers 78 authors. Coders received the poems in a fixed random order with author, title, dynasty, anthology membership and all lexicon output withheld, so the coding cannot be anchored on prior reputation.

Instrument and reliability. Each poem is rated on the five categories on a 0–3 ordinal scale (0 = absent, 1 = present once incidentally, 2 = clearly present, 3 = dominant), with the category definitions of §3 and two anchor examples per category agreed before coding began; both coders rated all 160 poems independently and no disagreement was resolved by discussion, so the reliability figures are not inflated by consensus. Agreement is reported as quadratic-weighted κ , ordinal Krippendorff α and ICC(2,1), and the gold index as the coder mean, exactly as in Eq. (2).

Why split-half here is contiguous, not odd–even. Per-poem reliability of the lexicon is computed by applying the lexicon separately to the first and second half of each poem’s characters and reporting the Spearman–Brown correction of the resulting correlation. Splitting by alternating characters would sever every bigram and every parallel pair (对仗), driving the correlation to zero for reasons that have nothing to do with the validity of a word-counting instrument; the estimate is therefore conservative in construction and its near-zero value should be read as a statement about the sparsity of poem-level word counts, not about the lexicon’s design. The same caution applies to $|\hat{\rho}|$: a lexicon that fires a handful of times in a 40-character poem cannot be reliable at that unit whatever its aggregate behaviour.

Gold-set sensitivity and per-author agreement. Dropping poems shorter than 28 characters ($n=150$), dropping the extreme 5% of lexicon scores ($n=147$) or dropping the longest poems ($n=150$) leaves the aggregate correlation at 0.43, 0.38 and 0.40 respectively, each with a 95% CI, confirming the aggregate correlation is not driven by any subset of poems. Per-author agreement is weaker: across the ten gold authors with at least three sampled poems the agreement is $\rho=0.13$ (95% CI $[-0.55, 0.69]$, $n=10$), indistinguishable from nothing at that sample size; we report it rather than search for the aggregation that flatters the instrument. The gold set is confined to *shi*; nothing here licenses the analyser for *ci* or prose.

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